

ABTAHI ISLAM

Software Developer | Full-Stack Web Development | AI/ML Engineer

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SUMMARY

Computer Science undergraduate at American International University - Bangladesh and Research Assistant at IdeaXLab who builds reliable, production-ready software across the full stack. Delivered a secure full-stack PHP/MySQL web portal with authenticated, role-based access; built REST APIs and CRUD services with Spring Boot, Hibernate, and Spring Security; and shipped multiple deployed front-end projects in HTML, CSS, and JavaScript. Complements this with applied AI/ML depth – explainable deep-learning pipelines for Alzheimer's and brain tumor MRI classification, and a production-grade agricultural chatbot achieving 99%+ model accuracy – backed by a peer-reviewed publication in IJECE (Q3) and a paper presented at an international BUTEX conference. Combines dependable engineering fundamentals, fast learning, and published research rigor with a consistent track record of shipping working software end to end.

WORK EXPERIENCE

JUL. 2026 – PRESENT

RESEARCH ASSISTANT – IDEAXLAB

Research & Innovation Laboratory

- Collaborate with researchers on experiment design, result analysis, methodology refinement, and preparation of research outputs for publication.

TECHNICAL SKILLS

Programming Languages:	Java, C++, C#, PHP, JavaScript, Python, R
Web & Backend Dev:	HTML5, CSS3, JavaScript, PHP, MySQL, REST APIs, Spring Boot, Hibernate/JPA, Spring Security
ML/DL Frameworks:	TensorFlow, Keras, Scikit-learn, XGBoost, LightGBM, CatBoost
NLP & Transformers:	Hugging Face, DistilBERT, n-gram modeling
Computer Vision:	CNN, transfer learning, medical image segmentation, Albumentations
Data & Analysis:	Pandas, NumPy, SPSS, R, statistical testing, SHAP, Optuna
Tools & Platforms:	Git, GitHub, XAMPP, Apache, Netlify, Google Colab, Jupyter

SOFTWARE & WEB DEVELOPMENT PROJECTS

MAY. 2026 – JUN. 2026

FTP Media Content System – ISP Media Portal (PHP/MySQL)

[🔗 Code](#)

- Built a full-stack PHP/MySQL web portal for an ISP media platform as part of a 4-person team; owned authentication – bcrypt password hashing, session-based login, 30-day "remember me" tokens (SHA-256 + HttpOnly cookies), CSRF protection, and role-aware navigation.

FEB. 2026 – PRESENT

Personal Portfolio Website

[🔗 Code](#) · [🔗 Live Demo](#)

- Designed and built a responsive personal portfolio in HTML, CSS, and vanilla JavaScript with a project showcase, certifications section, resume download, and custom cursor animation, deployed on Netlify and Vercel.

DEC. 2025 – PRESENT

Spring Boot & Hibernate Backend Development

[🔗 Code](#)

- Built REST APIs and CRUD services with Spring Boot 3, Spring Data JPA, and Hibernate; implemented Spring Security with bcrypt encryption, JDBC authentication, and role-based access control.

JUN. 2024

Electricity Billing System (Java Desktop App)

[🔗 Code](#)

- Developed a Java Swing desktop application automating customer registration, meter management, and bill calculation, with Bkash/bank payment integration and file-based data storage across 18 OOP-structured classes.

SEP. 2025

Presento – AI-Powered Multimodal Presentation Skill Assessment

[🔗 Code](#)

- This contains the full project documentation, following all phases of the Software Development Life Cycle (SDLC), except the coding/implementation phase. Here are Gantt chart milestones across *Analysis* → *Planning* → *Design* → *Implementation* → *Testing* → *Documentation*.

AI & MACHINE LEARNING PROJECTS

JUL. 2026 – PRESENT

AI-Assisted Early Detection of Alzheimer's Disease from Brain MRI

[Code](#)

- Built an explainable ML pipeline on 4,333 survey responses to predict 6 mental health outcomes. Optimized XGBoost with Optuna, achieving 75.6% test accuracy and 0.822 ROC-AUC, while integrating SHAP explainability and identifying a data leakage issue that had falsely inflated accuracy.

FEB. 2026 – MAR. 2026

A Comparative Study of Keras Models for Brain Tumor MRI Segmentation

[Code](#)

- Benchmarked 14 pretrained deep learning models for brain tumor MRI classification using a unified training pipeline. ResNet50 achieved 94.3% accuracy with 40% shorter training time than the best-performing EfficientNetB4.

JUN. 2026

AgriBot – Agricultural Advisory Chatbot for Farmers

[Code](#) · [Live Demo](#)

- Built a 4-layer hybrid AI chatbot using DistilBERT, Random Forest, rule-based reasoning, and a weather API. Developed ML models achieving up to 99.55% accuracy for crop prediction, 98.43% for plant disease detection, and 99.27% validation accuracy for DistilBERT intent classification with 2× faster inference than BERT.

PUBLICATIONS

APR. 2026

The ethics of AI technology in academic work: assessing the line between assistance and plagiarism

International Journal of Electrical and Computer Engineering (IJECE) (Q3) · [View Paper](#)

- Reviewed 24 FL oncology studies; compared model architectures and aggregation strategies across institutions.
- Identified privacy-utility trade-offs and proposed practical deployment guidelines for clinical settings.

EDUCATION HISTORY

JAN. 2023 – PRESENT

BACHELOR'S DEGREE, COMPUTER SCIENCE AND ENGINEERING

- American International University-Bangladesh | CGPA: 3.39

JAN. 2019 – NOV. 2021

HIGHER SECONDARY CERTIFICATE

- Ishwardi Government College | GPA: 5.00

JUL. 2018

Secondary School Certificate

- Govt. Sara Marwari Model School and College | GPA: 5.00

CERTIFICATION

MAY 2026 · [View Certificate](#)

Code to Cloud: Practical DevOps with AWS — Cloudly

JAN. 2026 · [View Certificate](#)

Spring Boot 4, Spring 7 & Hibernate for Beginners — Udemy

NOV. 2025 · [View Certificate](#)

Gemini Certified University Student — Google

MAY 2025 · [View Certificate](#)

Android Course Build 3 Applications from Scratch with Java — Udemy

JUL. 2025 · [View Certificate](#)

Java Complete Course Using Visual Studio Code — Udemy

FEB. 2026 · [View Certificate](#)

Agile Project Management — HP LIFE

REFERENCES

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